Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Material and test technologies for innovative, additively manufactured removable dental appliances			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (x), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric materials	Materials for medical applications	R&D/ Engineering Services	Testing / Inspection/ Analysis technique
Concept of Proposed Technology	Material characterization platform for: • Optimization of material and surface properties of additively manufactured removable dental appliances • Chemical and mechanical surface analytics to assess application behavior in terms of biofilm formation and optimal removal • Development of optimized material and surface properties and in relation to antimicrobial technologies • Chemical investigations for developing sustainability-oriented concepts			
Rationale for Joint R&D Project	This collaboration combines Fraunhofer's advanced expertise in material and surface characterization with the material and technology knowledge of Korean SMEs in additive manufacturing to accelerate the development, validation, and commercialization of innovative medical/dental devices.			
Key Words	• Dental (and medical) appliances • Innovative test (in vitro testing) and surface technologies • Microstructure, mechanical, chemical and optical investigations • Biofilm formation, plaque removal, cleaning technologies			

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	• Material and surface optimization with sustainability

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (9)
R&D Objectives	• Complex mechanical, microstructure and surface analyses for material development/optimization and further development of manufacturing technology • Development of a platform for material and prototype characterization / Development of test technologies for product application (simulation of "real-life" conditions) • Prototype data processing, screening (quality assurance) and in vitro testing			
R&D Target Specifications	• Complex material characterization investigations regarding additively manufactured dental materials (on correlation to process parameters) • Development of innovative surface technologies to reduce biofilm formation (e.g. using multi-species models), to optimize biofilm removal and to increase the material compatibility on additively manufactured removable dental appliances • Development of optimized surface compatible and antimicrobial cleansing technology • Chemical investigations for developing sustainability-oriented concepts			

R&D Plan

Phase	TRL goal	Research approach & methodologies	Key tasks	Expected milestones
1	Experimental stage (3-4)	Development of innovative surface technologies and tools	Development of innovative surface technologies Optimization of material surface properties Development of in vitro testing protocols	Verification of basic performance at a laboratory scale Evaluation of key performance of materials at a laboratory scale
2	Prototype finalization and evaluation (5-6)	Performance of in vitro testing using newly developed material surfaces	Microstructure and mechanical investigations Chemical surface analyzation Investigation of biofilm formation, plaque removal and antimicrobial properties	Production and performance evaluation of prototypes of finalized materials Performance evaluation at a pilot scale
3	Productization Stage (7-8)			Prototype certification and standardization
4	Commercialization Stage (9)			Commercialization

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Know-how provider and technology transfer to different industrial applications like dental and medical manufacturers • Working as a platform for the characterization of newly developed materials or surface technologies for industrial requirements • Understanding the production process and the optimal material design to identify relevant material characteristics • Evaluation and qualification of product behavior with regard to application (including all interactions occurring in 'real life')
Expected R&R of Korean SME Partner	• Specialized in the development of additively manufactured removable dental appliances • Offering innovative products in the field of dental appliances and other medical devices • Leading role due to its extensive expertise in developing advanced product systems and application • Using scientific material and technology data to support claims, marketing • Product expansion to strengthen market position

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• New, innovative surface technologies and in vitro test methods for additively manufactured removable dental appliances • Significantly improved understanding of the specific product properties to strengthen claims and marketing position • Improvement of in vitro testing methods to analyze and evaluate prototypes • Basis for the development and optimization of various dental applications (customized and product-specific) • By applying these new methodologies, Korean manufacturers of dental and medical appliances can significantly enhance their products and production technologies, thereby better competing with international competitors.

6. References

(1) Sarembe, S., M. Morawietz, A. Kiesow, E. Guritzky, and A. R. Hunt. "Cleaning Performance of Departure Cleansers on 3d-Generated Removable Oral Appliances." 2025 IADR General Session & Exhibition, Barcelona, ESP, International Association for Dental Research, June 25 to 28, 2025.

(2) Sarembe, S., M. Morawietz, A. Kiesow, E. Guritzky, and A. R. Hunt. "Material Compatibility of Denture Cleansers on Differently Manufactured Dental-Base Resins." 2025 IADR General Session & Exhibition, Barcelona, ESP, International Association for Dental Research, June 25 to 28, 2025.

(3) Sarembe S., Kiesow A., Guritzky E., Hunt A., Simkins J. “Material compatibility of denture cleansers with 3D-generated removable oral appliances” AADOCR/CADR Annual Meeting & Exhibition, New York City, USA, March 12 to 15, 2025

(4) Kiesow A, Sarembe S, Pizzey RL, Axe AS, Bradshaw DJ. Material compatibility and antimicrobial activity of consumer products commonly used to clean dentures. J Prosthet Dent. 2016 Feb;115(2):189-198.e8. doi: 10.1016/j.prosdent.2015.08.010. Epub 2015 Nov 3. PMID: 26545863.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization